

ISC Class XII Physics

Chapter 13 Nuclei

25 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Mass-Energy Relation
- II. Pair-Production and Annihilation of Matter
- III. Mass Defect and Binding Energy
- IV. Q-Value of Nuclear Reactions
- V. Nuclear Fission
- VI. Nuclear Fusion

Contents

1	I. Mass-Energy Relation	7
2	II. Pair-Production and Annihilation of Matter	9
3	III. Mass Defect and Binding Energy	11
4	IV. Q-Value of Nuclear Reactions	16
5	V. Nuclear Fission	17
6	VI. Nuclear Fusion	21

Index of Numericals

Q. No.	Topic / Statement Summary	Section
1	Percentage increase in Earth's mass due to absorbed solar energy	I
2	Energy equivalent of the rest mass of an electron (in joule)	I
3	Rest-mass energy of an electron in MeV ★	I
4	Energy equivalent of 1 gram mass, in electron-volt	I
5	Minimum photon energy and maximum wavelength for e^- - e^+ pair production	II
6	Energy produced in proton-antiproton annihilation	II
7	Wavelength of photons from proton-antiproton annihilation	II
8	Energy liberated in 1 kg fusion reaction (0.3% mass defect)	III
9	Energy to remove a neutron from ${}_8O^{17}$ using BE/nucleon values ★	III
10	Energy carried by each α -particle in ${}_3Li^7 + {}_1H^1$ reaction	III
11	Value of n : energy liberated ($n \times 10^7$ kWh) from 0.4 g mass defect	III
12	Binding energy of the ${}_8O^{16}$ nucleus ★	III

Q. No.	Topic / Statement Summary	Section
13	Energy released forming ${}_3\text{Li}^7$ from 4 neutrons and 3 protons	III
14	Energy needed to remove a neutron from ${}_{13}\text{Al}^{27}$	III
15	Energy released: 100 g of ${}_3\text{Li}^7$ converted to ${}_2\text{He}^4$ by proton bombardment	III
16	Disintegration energy Q for symmetric fission of an $A = 240$ nucleus ★	IV
17	Energy from 0.5 g mass decay in uranium fission (in MeV and kWh)	V
18	Energy and electricity from 0.10% mass loss in fission of 1 kg substance	V
19	Energy released on fissioning 100 g of ${}_{92}\text{U}^{235}$ ★	V
20	Fission rate and U^{235} consumption in a 32,000 kW reactor ★	V
21	Energy (in kWh) from fusion of 1 g hydrogen at 5% generator efficiency ★	V
22	Fission rate for 1 kW power output; total energy from 0.5 kg U^{235} fission	V
23	Energy evolved in triple-alpha fusion forming ${}_6\text{C}^{12}$	VI
24	Energy released: four ${}_1\text{H}^1$ nuclei fuse to form helium with 2 positrons ★	VI
25	Q-value of the fusion reaction ${}_1\text{H}^2 + {}_1\text{H}^3 \rightarrow {}_2\text{He}^4 + {}_0\text{n}^1$ ★	VI

Formula Sheet

Standard Constants used in this Chapter:

$$m_p = 1.00728 \text{ u} \quad m_n = 1.00867 \text{ u} \quad m_e = 0.00055 \text{ u}$$

$$h = 6.6 \times 10^{-34} \text{ J-s} \quad c = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$1 \text{ MeV} = 1.60 \times 10^{-13} \text{ J} \quad 1 \text{ u} = 931.5 \text{ MeV (i.e. } 1 \text{ u} \times c^2 = 931.5 \text{ MeV)}$$

I. Mass-Energy Relation

Mass-Energy Equivalence

$$E = mc^2$$

E = energy released/absorbed (J); m = mass converted (kg); c = speed of light

Percentage Change in Mass

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100$$

Δm = mass gained/lost; M = original mass of body

II. Pair-Production and Annihilation of Matter

Minimum Energy for Pair Production

$$E_{min} = 2m_e c^2$$

m_e = rest mass of electron (equal to rest mass of positron)

Photon Energy-Wavelength Relation

$$E = \frac{hc}{\lambda}$$

h = Planck's constant; λ = wavelength of the photon

Particle-Antiparticle Annihilation Energy

$$E = 2mc^2$$

m = rest mass of each particle (e.g. proton and antiproton)

III. Mass Defect and Binding Energy

Mass Defect

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{nucleus}$$

Z = atomic number; A = mass number; $M_{nucleus}$ = actual nuclear mass

Binding Energy

$$BE = \Delta m \text{ (in u)} \times 931.5 \text{ MeV}$$

Δm expressed in atomic mass units (u)

Binding Energy per Nucleon

$$\frac{BE}{A}$$

Measure of nuclear stability; higher value $\Rightarrow$ more stable nucleus

IV. Q-Value of Nuclear Reactions

Q-Value

$$Q = (\Sigma m_{reactants} - \Sigma m_{products}) \times 931.5 \text{ MeV}$$

Positive Q = energy released (exothermic); Negative Q = energy absorbed

V. Nuclear Fission

Fission Energy

$$E = \Delta mc^2$$

Δm = mass defect in the fission process

Energy-kWh Conversion

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Used to express large nuclear energies in electrical units

Fission Rate

$$\frac{dN}{dt} = \frac{P}{E_{fission}}$$

P = power output (W); $E_{fission}$ = energy released per fission (J)

VI. Nuclear Fusion

Fusion Energy

$$E = \Delta mc^2$$

Δm = mass defect between fusing nuclei and product nucleus

Useful Energy with Efficiency

$$E_{useful} = \eta \times E_{total}$$

η = fractional efficiency of the generator/reactor

I. Mass-Energy Relation

Question 1

Earth absorbs 10^{22} joule energy everyday from the sun. What will be the percentage increase in the earth's mass (6.0×10^{24} kg) due to this?

Given: $E = 10^{22}$ J, $M = 6.0 \times 10^{24}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = \Delta mc^2 \Rightarrow \% \text{ increase} = \frac{\Delta m}{M} \times 100$

Solution:

$$\Delta m = \frac{E}{c^2} = \frac{10^{22}}{(3 \times 10^8)^2} = \frac{10^{22}}{9 \times 10^{16}} = 1.111 \times 10^5 \text{ kg}$$

$$\% \text{ increase} = \frac{\Delta m}{M} \times 100 = \frac{1.111 \times 10^5}{6.0 \times 10^{24}} \times 100 = 1.85 \times 10^{-18} \%$$

Final Answer

Percentage increase in Earth's mass $\approx 1.85 \times 10^{-18} \%$

Common Mistake: Don't forget to multiply by 100 for a percentage, and keep careful track of the very large/small powers of ten – a single exponent slip changes the answer completely.

Question 2

Work out the energy equivalence of the rest-mass of electron (9.1×10^{-31} kg).

Given: $m_e = 9.1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = mc^2$

Solution:

$$E = 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 9.1 \times 10^{-31} \times 9 \times 10^{16} = 8.19 \times 10^{-14} \text{ J}$$

Final Answer

$E \approx 8.2 \times 10^{-14}$ J

Question 3

Find the rest-mass energy of electron in MeV.

★ Likely Exam Question

Given: $E = 8.2 \times 10^{-14}$ J (from Q2), $1 \text{ MeV} = 1.6 \times 10^{-13}$ J

Formula: $E(\text{MeV}) = \frac{E(\text{J})}{1.6 \times 10^{-13}}$

Solution:

$$E = \frac{8.2 \times 10^{-14}}{1.6 \times 10^{-13}} = 0.5125 \text{ MeV}$$

Final Answer

Rest-mass energy of electron ≈ 0.5125 MeV

Common Mistake: This value (≈ 0.511 MeV) is a standard constant used repeatedly in pair-production questions – memorise it.

Question 4

What is the equivalent energy in electron-volt of 1 gram mass?

Given: $m = 1 \text{ g} = 10^{-3} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Formula: $E = mc^2$

Solution:

$$E = 10^{-3} \times (3 \times 10^8)^2 = 10^{-3} \times 9 \times 10^{16} = 9 \times 10^{13} \text{ J}$$

$$E(\text{eV}) = \frac{9 \times 10^{13}}{1.6 \times 10^{-19}} = 5.625 \times 10^{32} \text{ eV}$$

Final Answer

$$E \approx 5.625 \times 10^{32} \text{ eV}$$

Common Mistake: Use $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ here (not $1.6 \times 10^{-13} \text{ J}$ for MeV) – mixing these two conversion factors is a very common slip.

II. Pair-Production and Annihilation of Matter

Question 5

What should be the minimum energy of a γ -ray photon for the production of an electron-positron pair? What can be the maximum wavelength? The rest-mass of electron (also of positron) is 9.1×10^{-31} kg.

Given: $m_e = 9.1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m s⁻¹, $h = 6.6 \times 10^{-34}$ J-s

Formula: $E_{min} = 2m_e c^2$ $\lambda_{max} = \frac{hc}{E_{min}}$

Solution:

$$E_{min} = 2 \times 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 2 \times 8.19 \times 10^{-14} = 1.638 \times 10^{-13} \text{ J}$$

$$E_{min}(\text{MeV}) = \frac{1.638 \times 10^{-13}}{1.6 \times 10^{-13}} = 1.02 \text{ MeV}$$

$$\lambda_{max} = \frac{hc}{E_{min}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.638 \times 10^{-13}} = 1.209 \times 10^{-12} \text{ m} = 0.012 \text{ \AA}$$

Final Answer

$$E_{min} \approx 1.02 \text{ MeV}, \quad \lambda_{max} \approx 0.012 \text{ \AA}$$

Common Mistake: Pair production creates *two* particles (electron + positron), so the threshold energy is $2m_e c^2$, not $m_e c^2$ – a very common exam trap.

Question 6

Calculate in joule the energy produced by the interaction of a proton and an antiproton. The mass of proton is 1.6725×10^{-27} kg.

Given: $m_p = 1.6725 \times 10^{-27}$ kg, $c = 3 \times 10^8$ m s⁻¹

Formula: $E = 2mc^2$

Solution:

$$E = 2 \times 1.6725 \times 10^{-27} \times (3 \times 10^8)^2 = 2 \times 1.6725 \times 10^{-27} \times 9 \times 10^{16}$$

$$E = 3.0 \times 10^{-10} \text{ J}$$

Final Answer

$$E \approx 3.0 \times 10^{-10} \text{ J}$$

Common Mistake: Both the proton *and* the antiproton contribute mass to the annihilation – omitting the factor of 2 halves the correct answer.

Question 7

A proton and an antiproton annihilate into two photons of same frequency. What is the wavelength of the photons produced? The mass of proton is $1.67 \times 10^{-27} \text{ kg}$.

Given: $m_p = 1.67 \times 10^{-27} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $h = 6.6 \times 10^{-34} \text{ J-s}$

Formula: $E_{total} = 2mc^2$, $E_{photon} = \frac{E_{total}}{2}$, $\lambda = \frac{hc}{E_{photon}}$

Solution:

$$E_{total} = 2 \times 1.67 \times 10^{-27} \times 9 \times 10^{16} = 3.006 \times 10^{-10} \text{ J}$$

Since the two photons carry the total energy equally,

$$E_{photon} = \frac{3.006 \times 10^{-10}}{2} = 1.503 \times 10^{-10} \text{ J}$$

$$\lambda = \frac{hc}{E_{photon}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.503 \times 10^{-10}} = 1.3 \times 10^{-15} \text{ m}$$

Final Answer

$$\lambda \approx 1.3 \times 10^{-15} \text{ m}$$

Common Mistake: Divide the total annihilation energy by 2 before finding λ – each photon carries only half the total energy released.

III. Mass Defect and Binding Energy

Question 8

The mass-defect in a nuclear fusion reaction is 0.3 per cent. What amount of energy will be liberated in 1 kg fusion reaction?

Given: $M = 1$ kg, mass defect = 0.3% of M , $c = 3 \times 10^8$ m s⁻¹

Formula: $E = \Delta mc^2$

Solution:

$$\Delta m = 0.3\% \times 1 \text{ kg} = 0.003 \text{ kg}$$

$$E = 0.003 \times (3 \times 10^8)^2 = 0.003 \times 9 \times 10^{16} = 2.7 \times 10^{14} \text{ J}$$

Final Answer

$$E = 2.7 \times 10^{14} \text{ J}$$

Question 9

The binding energy per nucleon of ${}_8\text{O}^{16}$ is 7.97 MeV and that of ${}_8\text{O}^{17}$ is 7.75 MeV. Find the energy needed to remove a neutron from ${}_8\text{O}^{17}$.

★ Likely Exam Question

Given: $(BE/A)_{\text{O}^{16}} = 7.97$ MeV, $(BE/A)_{\text{O}^{17}} = 7.75$ MeV

Formula: Energy needed = $BE(\text{O}^{17}) - BE(\text{O}^{16})$

Solution:

$$BE(\text{O}^{16}) = 16 \times 7.97 = 127.52 \text{ MeV}$$

$$BE(\text{O}^{17}) = 17 \times 7.75 = 131.75 \text{ MeV}$$

$$\text{Energy needed} = 131.75 - 127.52 = 4.23 \text{ MeV}$$

Final Answer

Energy needed to remove a neutron = 4.23 MeV

Common Mistake: Always convert BE/nucleon to *total* binding energy (multiply by mass number A) before subtracting – subtracting the per-nucleon values directly gives a wrong answer.

Question 10

Bombardment of lithium with protons give rise to the following reaction:



The atomic masses of lithium, hydrogen and helium are 7.016 u, 1.008 u and 4.004 u respectively. Find the energy carried by each α -particle. Given: 1 u = 931 MeV.

Given: $m(\text{Li}^7) = 7.016 \text{ u}$, $m(\text{H}^1) = 1.008 \text{ u}$, $m(\text{He}^4) = 4.004 \text{ u}$, 1 u = 931 MeV

Formula: $\Delta m = [m(\text{Li}^7) + m(\text{H}^1)] - 2m(\text{He}^4)$, $E_{\text{total}} = \Delta m \times 931 \text{ MeV}$

Solution:

$$\Delta m = (7.016 + 1.008) - 2(4.004) = 8.024 - 8.008 = 0.016 \text{ u}$$

$$E_{\text{total}} = 0.016 \times 931 = 14.896 \text{ MeV}$$

Since two identical α -particles are produced, each carries half the total energy:

$$E_{\alpha} = \frac{14.896}{2} = 7.448 \text{ MeV}$$

Final Answer

Energy carried by each α -particle = 7.448 MeV

Common Mistake: Use the 1 u = 931 MeV value exactly as given in the question (not 931.5 MeV) so the final figure matches the expected answer.

Question 11

The mass defect in a particular reaction is 0.4 g. The amount of energy liberated is $n \times 10^7$ kWh. What is the value of n here? (speed of light = 3×10^8 m/s)

Given: $\Delta m = 0.4 \text{ g} = 0.4 \times 10^{-3} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$

Formula: $E = \Delta mc^2$, 1 kWh = $3.6 \times 10^6 \text{ J}$

Solution:

$$E = (0.4 \times 10^{-3}) \times (3 \times 10^8)^2 = 0.4 \times 10^{-3} \times 9 \times 10^{16} = 3.6 \times 10^{13} \text{ J}$$

$$E(\text{kWh}) = \frac{3.6 \times 10^{13}}{3.6 \times 10^6} = 1 \times 10^7 \text{ kWh}$$

Comparing with $n \times 10^7$ kWh,

$$n = 1$$

Final Answer

$n = 1$

Question 12

Determine the binding energy of the nucleus of ${}_8\text{O}^{16}$. Given: mass of proton = 1.007825 u, mass of neutron = 1.008665 u and atomic mass of ${}_8\text{O}^{16} = 15.994915$ u.

★ Likely Exam Question

Given: $Z = 8$, $A = 16$, $m_p = 1.007825$ u, $m_n = 1.008665$ u, $M = 15.994915$ u

Formula: $\Delta m = Zm_p + (A - Z)m_n - M$, $BE = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 8(1.007825) + 8(1.008665) - 15.994915$$

$$\Delta m = 8.062600 + 8.069320 - 15.994915 = 16.131920 - 15.994915 = 0.137005 \text{ u}$$

$$BE = 0.137005 \times 931.5 = 127.616 \text{ MeV}$$

Final Answer

Binding energy of ${}_8\text{O}^{16} = 127.616$ MeV

Common Mistake: Here proton and neutron masses (not the hydrogen atom mass) are given directly – use them as they are; don't further subtract electron mass separately.

Question 13

If we form a lithium nucleus (${}_3\text{Li}^7$) by 4 neutrons and 3 protons, then how much energy will be released? The mass of lithium nucleus is 7.01436 u.

Given: 4 neutrons, 3 protons, $m_n = 1.00867$ u, $m_p = 1.00728$ u, $M(\text{Li}^7) = 7.01436$ u

Formula: $\Delta m = 4m_n + 3m_p - M$, $E = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 4(1.00867) + 3(1.00728) - 7.01436$$

$$\Delta m = 4.03468 + 3.02184 - 7.01436 = 7.05652 - 7.01436 = 0.04216 \text{ u}$$

$$E = 0.04216 \times 931.5 = 39.27 \text{ MeV}$$

Final Answer

Energy released = 39.27 MeV

Question 14

How much energy is needed to remove a neutron from the nucleus of ${}_{13}\text{Al}^{27}$? Given: mass of ${}_{13}\text{Al}^{27}$ atom = 26.981541 u, mass of ${}_{13}\text{Al}^{26}$ atom = 25.986895 u, mass of neutron = 1.008665 u.

Given: $M(\text{Al}^{27}) = 26.981541$ u, $M(\text{Al}^{26}) = 25.986895$ u, $m_n = 1.008665$ u

Formula: $E = [M(\text{Al}^{26}) + m_n - M(\text{Al}^{27})] \times 931.5$ MeV

Solution:

$$M(\text{Al}^{26}) + m_n = 25.986895 + 1.008665 = 26.995560 \text{ u}$$

$$\Delta m = 26.995560 - 26.981541 = 0.014019 \text{ u}$$

$$E = 0.014019 \times 931.5 = 13.06 \text{ MeV}$$

Final Answer

Energy needed to remove a neutron = 13.06 MeV

Common Mistake: Set up the mass difference as (products – reactant is reversed here) – the energy *absorbed* equals [mass of fragments] – [mass of original nucleus], the opposite sign convention from a normal exothermic reaction.

Question 15

100 g of ${}_{3}\text{Li}^7$ is converted into ${}_{2}\text{He}^4$ by proton bombardment. Calculate the energy released. Given: mass of ${}_{3}\text{Li}^7$ atom = 7.016 u, mass of ${}_{2}\text{He}^4$ atom = 4.004 u, mass of ${}_{1}\text{H}^1$ atom = 1.008 u. The Avogadro number is $6.02 \times 10^{23} \text{ mol}^{-1}$.

Given: $m = 100$ g of Li^7 , $M(\text{Li}^7) = 7.016$ u, $M(\text{He}^4) = 4.004$ u, $M(\text{H}^1) = 1.008$ u, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Formula: $\Delta m = [M(\text{Li}^7) + M(\text{H}^1)] - 2M(\text{He}^4)$, $E_{\text{total}} = N \times \Delta m \times 931.5$ MeV

Solution:

Mass defect per reaction:

$$\Delta m = (7.016 + 1.008) - 2(4.004) = 8.024 - 8.008 = 0.016 \text{ u}$$

Energy per reaction:

$$E_1 = 0.016 \times 931.5 = 14.904 \text{ MeV} = 14.904 \times 1.6 \times 10^{-13} \text{ J} = 2.385 \times 10^{-12} \text{ J}$$

Number of Li^7 atoms in 100 g (using atomic mass ≈ 7 g/mol):

$$N = \frac{100}{7} \times 6.02 \times 10^{23} = 8.6 \times 10^{24} \text{ atoms}$$

Total energy released:

$$E_{total} = N \times E_1 = 8.6 \times 10^{24} \times 2.385 \times 10^{-12} \approx 2 \times 10^{13} \text{ J}$$

Final Answer

Total energy released $\approx 2 \times 10^{13} \text{ J}$

Common Mistake: Find the number of moles of Li^7 (mass/molar mass) first, then multiply by N_A – do not multiply 100 g directly by N_A .

IV. Q-Value of Nuclear Reactions

Question 16

Calculate the disintegration energy Q when a heavy nucleus of mass number $A = 240$ with binding energy per nucleon 7.6 MeV is split into two equal fragments of mass number $A' = 120$, each with $\text{BE/nucleon } 8.5 \text{ MeV}$.

★ Likely Exam Question

Given: $A = 240$, $(\text{BE}/A)_{\text{initial}} = 7.6 \text{ MeV}$, $A' = 120$ (two fragments), $(\text{BE}/A)_{\text{final}} = 8.5 \text{ MeV}$

Formula: $Q = BE_{\text{final}} - BE_{\text{initial}}$

Solution:

$$BE_{\text{initial}} = 240 \times 7.6 = 1824 \text{ MeV}$$

$$BE_{\text{final}} = 2 \times (120 \times 8.5) = 2 \times 1020 = 2040 \text{ MeV}$$

$$Q = 2040 - 1824 = 216 \text{ MeV}$$

Final Answer

Disintegration energy $Q = 216 \text{ MeV}$

Common Mistake: Remember there are *two* product fragments – forgetting to double the final binding energy is a very common slip in fission Q -value problems.

V. Nuclear Fission

Question 17

0.5 g mass is decayed in the fission of a part of uranium. How much energy will be obtained by it? Express the result in MeV and in kWh.

Given: $\Delta m = 0.5 \text{ g} = 0.5 \times 10^{-3} \text{ kg}$, $c = 3 \times 10^8 \text{ m s}^{-1}$

Formula: $E = \Delta mc^2$, $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Solution:

$$E = (0.5 \times 10^{-3}) \times 9 \times 10^{16} = 4.5 \times 10^{13} \text{ J}$$

$$E(\text{MeV}) = \frac{4.5 \times 10^{13}}{1.6 \times 10^{-13}} = 2.8 \times 10^{26} \text{ MeV}$$

$$E(\text{kWh}) = \frac{4.5 \times 10^{13}}{3.6 \times 10^6} = 1.25 \times 10^7 \text{ kWh}$$

Final Answer

$$E \approx 2.8 \times 10^{26} \text{ MeV} = 1.25 \times 10^7 \text{ kWh}$$

Question 18

If the loss of mass in some fission process is 0.10%, then how much energy will be obtained by the fission of 1 kg substance? How many kilowatt-hours of electricity can be obtained by it?

Given: $M = 1 \text{ kg}$, mass loss = 0.10% of M

Formula: $E = \Delta mc^2$, $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Solution:

$$\Delta m = 0.10\% \times 1 \text{ kg} = 1 \times 10^{-3} \text{ kg}$$

$$E = (1 \times 10^{-3}) \times 9 \times 10^{16} = 9 \times 10^{13} \text{ J}$$

$$E(\text{kWh}) = \frac{9 \times 10^{13}}{3.6 \times 10^6} = 2.5 \times 10^7 \text{ kWh}$$

Final Answer

$$E = 9 \times 10^{13} \text{ J} = 2.5 \times 10^7 \text{ kWh}$$

Question 19

The energy released per fission of ${}_{92}\text{U}^{235}$ nucleus is 200 MeV. Compute the energy released in joule if 100 g of ${}_{92}\text{U}^{235}$ is fissioned. Avogadro's number is $6.02 \times 10^{23} \text{ mole}^{-1}$.

★ Likely Exam Question

Given: mass of $\text{U}^{235} = 100 \text{ g}$, $E_{\text{fission}} = 200 \text{ MeV}$, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

Formula: $N = \frac{\text{mass}}{\text{molar mass}} \times N_A$, $E_{\text{total}} = N \times E_{\text{fission}}$

Solution:

Number of moles of U^{235} :

$$n = \frac{100}{235} = 0.4255 \text{ mol}$$

Number of atoms:

$$N = 0.4255 \times 6.02 \times 10^{23} = 2.562 \times 10^{23} \text{ atoms}$$

Total energy released:

$$E_{\text{total}} = N \times E_{\text{fission}} = 2.562 \times 10^{23} \times 200 \text{ MeV} = 5.124 \times 10^{25} \text{ MeV}$$

$$E_{\text{total}}(\text{J}) = 5.124 \times 10^{25} \times 1.6 \times 10^{-13} = 8.2 \times 10^{12} \text{ J}$$

Final Answer

Energy released $\approx 8.2 \times 10^{12} \text{ J}$

Question 20

A U^{235} reactor is producing nuclear energy at a rate of 32,000 kW. How many nuclei of U^{235} are fissioned per second? How much U^{235} would be consumed in 1000 hours? Assume an average energy of 200 MeV released per fission. Take Avogadro's number as $6 \times 10^{23} \text{ mol}^{-1}$.

★ Likely Exam Question

Given: $P = 32,000 \text{ kW} = 3.2 \times 10^7 \text{ W}$, $E_{\text{fission}} = 200 \text{ MeV}$, $t = 1000 \text{ h}$, $N_A = 6 \times 10^{23} \text{ mol}^{-1}$

Formula: $\frac{dN}{dt} = \frac{P}{E_{\text{fission}}}$, $\text{mass consumed} = \frac{N_{\text{total}}}{N_A} \times 235 \text{ g}$

Solution:

Energy released per fission:

$$E_{\text{fission}} = 200 \times 1.6 \times 10^{-13} = 3.2 \times 10^{-11} \text{ J}$$

Fissions per second:

$$\frac{dN}{dt} = \frac{3.2 \times 10^7}{3.2 \times 10^{-11}} = 1.0 \times 10^{18} \text{ fissions/s}$$

Total fissions in 1000 hours ($t = 1000 \times 3600 = 3.6 \times 10^6$ s):

$$N_{\text{total}} = 1.0 \times 10^{18} \times 3.6 \times 10^6 = 3.6 \times 10^{24} \text{ nuclei}$$

Moles of U^{235} consumed:

$$n = \frac{3.6 \times 10^{24}}{6 \times 10^{23}} = 6 \text{ mol}$$

Mass consumed:

$$m = 6 \times 235 = 1410 \text{ g} = 1.41 \text{ kg}$$

Final Answer

Fission rate = 1.0×10^{18} nuclei/sec; mass of U^{235} consumed in 1000 h = 1.41 kg

Common Mistake: Convert 1000 hours into seconds ($\times 3600$) before finding total fissions – mixing time units is the most common error in reactor-rate problems.

Question 21

In nuclear fusion, 0.993 g helium is formed from 1 g hydrogen. If the efficiency of the generator be 5%, then calculate the energy obtained in kilowatt-hour (kWh).

★ Likely Exam Question

Given: mass of hydrogen = 1 g, mass of helium formed = 0.993 g, efficiency $\eta = 5\% = 0.05$

Formula: $\Delta m = m_H - m_{He}$, $E = \Delta mc^2$, $E_{\text{useful}} = \eta \times E$

Solution:

$$\Delta m = 1 - 0.993 = 0.007 \text{ g} = 7 \times 10^{-6} \text{ kg}$$

$$E = (7 \times 10^{-6}) \times 9 \times 10^{16} = 6.3 \times 10^{11} \text{ J}$$

$$E_{\text{useful}} = 0.05 \times 6.3 \times 10^{11} = 3.15 \times 10^{10} \text{ J}$$

$$E_{\text{useful}}(\text{kWh}) = \frac{3.15 \times 10^{10}}{3.6 \times 10^6} = 8.75 \times 10^3 \text{ kWh}$$

Final Answer

Energy obtained $\approx 8.75 \times 10^3$ kWh

Common Mistake: Apply the 5% efficiency factor to the total energy released – do not apply it to the mass defect or skip it altogether.

Question 22

If 200 MeV energy is released in the fission of a single nucleus of ${}_{92}\text{U}^{235}$, how many fissions must occur in one second to produce a power of 1 kW? Also compute the energy released in the complete fission of 0.5 kg of U^{235} . Take Avogadro's number $= 6.02 \times 10^{26} \text{ kmol}^{-1}$.

Given: $E_{\text{fission}} = 200 \text{ MeV}$, $P = 1 \text{ kW} = 1000 \text{ W}$, $\text{mass} = 0.5 \text{ kg}$, $N_A = 6.02 \times 10^{26} \text{ kmol}^{-1}$

Formula: $\frac{dN}{dt} = \frac{P}{E_{\text{fission}}}$, $E_{\text{total}} = \frac{\text{mass}}{\text{molar mass}} \times N_A \times E_{\text{fission}}$

Solution:

Energy per fission:

$$E_{\text{fission}} = 200 \times 1.6 \times 10^{-13} = 3.2 \times 10^{-11} \text{ J}$$

Fissions per second for 1 kW:

$$\frac{dN}{dt} = \frac{1000}{3.2 \times 10^{-11}} = 3.125 \times 10^{13} \text{ fissions/s}$$

For 0.5 kg of U^{235} (molar mass 235 kg/kmol):

$$n = \frac{0.5}{235} = 2.128 \times 10^{-3} \text{ kmol}$$

$$N = 2.128 \times 10^{-3} \times 6.02 \times 10^{26} = 1.281 \times 10^{24} \text{ atoms}$$

$$E_{\text{total}} = 1.281 \times 10^{24} \times 200 \text{ MeV} = 2.56 \times 10^{26} \text{ MeV}$$

Final Answer

Fission rate $= 3.125 \times 10^{13} \text{ /sec}$; energy from complete fission of 0.5 kg $\text{U}^{235} = 2.56 \times 10^{26} \text{ MeV}$

VI. Nuclear Fusion

Question 23

In a star, three α -particles (${}_2\text{He}^4$ nuclei) fuse together to form a ${}_6\text{C}^{12}$ nucleus. Compute the energy evolved. The masses of ${}_2\text{He}^4$ atom and ${}_6\text{C}^{12}$ atom are 4.002604 u and 12.000000 u respectively.

Given: $3 \times {}_2\text{He}^4 \rightarrow {}_6\text{C}^{12}$, $M(\text{He}^4) = 4.002604$ u, $M(\text{C}^{12}) = 12.000000$ u

Formula: $\Delta m = 3M(\text{He}^4) - M(\text{C}^{12})$, $E = \Delta m \times 931.5$ MeV

Solution:

$$\Delta m = 3(4.002604) - 12.000000 = 12.007812 - 12.000000 = 0.007812 \text{ u}$$

$$E = 0.007812 \times 931.5 = 7.277 \text{ MeV}$$

Final Answer

Energy evolved = 7.277 MeV

Question 24

Four hydrogen nuclei fuse together to form a helium nucleus with emission of two positrons. Calculate the energy released in the reaction. Given: mass of ${}_1\text{H}^1$ atom = 1.007825 u, mass of ${}_2\text{He}^4$ atom = 4.002603 u, mass of positron = 0.000549 u.

★ Likely Exam Question

Given: $4{}_1\text{H}^1 \rightarrow {}_2\text{He}^4 + 2e^+$, $M(\text{H}^1) = 1.007825$ u, $M(\text{He}^4) = 4.002603$ u, $m_{e^+} = 0.000549$ u

Formula: $\Delta m = 4M(\text{H}^1) - [M(\text{He}^4) + 2m_{e^+}]$, $E = \Delta m \times 931.5$ MeV

Solution:

$$4M(\text{H}^1) = 4 \times 1.007825 = 4.031300 \text{ u}$$

$$M(\text{He}^4) + 2m_{e^+} = 4.002603 + 2(0.000549) = 4.002603 + 0.001098 = 4.003701 \text{ u}$$

$$\Delta m = 4.031300 - 4.003701 = 0.027599 \text{ u}$$

$$E = 0.027599 \times 931.5 = 25.71 \text{ MeV}$$

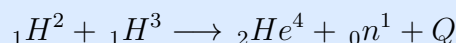
Final Answer

Energy released = 25.71 MeV

Common Mistake: The two emitted positrons carry mass too – their combined mass must be added to the helium mass on the product side before subtracting; leaving them out overstates the mass defect.

Question 25

The following equation represents a fusion reaction:



where Q is the energy released. Given: mass of ${}_1H^2$ atom = 2.014102 u, mass of ${}_1H^3$ atom = 3.016050 u, mass of ${}_2He^4$ atom = 4.002603 u and mass of neutron ${}_0n^1 = 1.008665$ u. Calculate the value of Q .

★ Likely Exam Question

Given: $M(H^2) = 2.014102$ u, $M(H^3) = 3.016050$ u, $M(He^4) = 4.002603$ u, $m_n = 1.008665$ u

Formula: $\Delta m = [M(H^2) + M(H^3)] - [M(He^4) + m_n]$, $Q = \Delta m \times 931.5$ MeV

Solution:

$$M(H^2) + M(H^3) = 2.014102 + 3.016050 = 5.030152 \text{ u}$$

$$M(He^4) + m_n = 4.002603 + 1.008665 = 5.011268 \text{ u}$$

$$\Delta m = 5.030152 - 5.011268 = 0.018884 \text{ u}$$

$$Q = 0.018884 \times 931.5 = 17.59 \text{ MeV}$$

Final Answer

$Q = 17.59$ MeV

Common Mistake: This is the classic D-T fusion reaction that powers fusion reactors – always subtract product mass from reactant mass ($\Sigma m_{\text{reactants}} - \Sigma m_{\text{products}}$) to keep Q positive for an exothermic reaction.